Muhammad Murtaza Tantry

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Employment History

2023 – 2026 **Teaching Assistant,** *Vellore Institute of Technology, Vellore*
Subjects: Calculus, Linear Algebra, and Python.

2021 – 2023 **Subject Matter Expert - Advanced Mathematics,** *Chegg.com*

Education

2023 – **Ph.D.,** *Vellore Institute of Technology, Vellore*
Thesis title: *Physics-Informed Neural Networks for Simulating Boundary Layer Flow of Viscoelastic Fluids*
Advisor: Dr. Padigepati Naveen

2018 – 2020 **M.Sc. Mathematics,** *University of Hyderabad*

2015 – 2018 **B.Sc. (Hons.) Mathematics,** *Jamia Millia Islamia*

Awards and Honors

- **GATE Mathematics**. AIR-115.

Research Interests

- Physics-Informed Neural Networks (PINNs); Scientific machine learning; Numerical methods for nonlinear differential equations; Boundary layer flow; Computational fluid dynamics

Research Publications

Journal Articles

1. **M. M. Tantry** and P. Naveen, "Adaptive loss balanced physics-informed neural networks for non-similar flow of maxwell fluid: A deep learning framework," *Journal of Thermal Analysis and Calorimetry*, 2026, Accepted for publication.
2. **M. M. Tantry** and P. Naveen, "Thomson and Troian slip condition for radiating second-grade nanofluid flow with shape effects of gold nanoparticles," *Numerical Heat Transfer, Part A: Applications*, vol. 87, no. 1, p. 2 527 403, 2026.
3. P. Naveen, **M. M. Tantry**, P. Sathya, and S. Sindhuja, "Artificial neural network modelling for Casson hybrid nanofluid flow with entropy generation analysis," *Multiscale and Multidisciplinary Modeling, Experiments and Design*, vol. 8, no. 6, p. 288, 2025.
4. **M. M. Tantry** and P. Naveen, "Aggregation kinematics on gold nanoparticles in radiating viscoelastic second-grade fluid flow with Thomson and Troian velocity slip condition," *Journal of Thermal Analysis and Calorimetry*, vol. 150, no. 12, pp. 9279–9291, 2025.

Manuscripts Communicated

- **M. M. Tantry** and P. Naveen, "A Deep Learning Approach with PINNs for Simulating Second-Grade Hybrid Nanofluid Flow over a Radially Extending Sheet", **under review**
- **M. M. Tantry** and P. Naveen, "Development of Physics-informed Neural Network Solver for Radiative Flow of Jeffrey Fluid with Thompson-Troian Slip Condition: A Deep Learning Approach", submitted [2026].
- **M. M. Tantry** and P. Naveen, "Physics-informed Neural Networks Framework for Non-similar Solutions of Maxwell Fluid Flow over a Thin Needle: A Deep Learning Approach", submitted [2026].

Talks and Presentations

[2026] **PhD Colloquium,** Department of Mathematics, VIT Vellore - Presented four research problems on Physics-Informed Neural Networks for viscoelastic boundary layer flows.

Skills

Languages	Strong reading, writing and speaking competencies for English, Urdu, Kashmiri.
Programming	Python, MATLAB, LaTeX
ML & Scientific Computing	PyTorch, TensorFlow, DeepXDE (physics-informed neural networks), SciPy
HPC	SLURM job scheduling and array jobs, Conda environment management
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Workshops and Training

2024 **NCMW – Numerical Methods for Partial Differential Equations,** Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam.

2023 **5-Day International Workshop on Recent Advances in Differential Equations and Applications (IWRADEA),** VIT-AP University, Andhra Pradesh. 27–31 October 2023.

References

Dr. Padigepati Naveen, Assistant Professor, Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Vellore.